Paing Hein Htet

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EDUCATION

University College London, MEng Biomedical Engineering *Sept 2025 – Jun 2029*

First Class (year one). **Kaggle bronze medal:** 549th of 10,148, Kaggriculture 2026. IEEE and IET student member.
A-Levels (2025): A* Maths, Further Maths, Physics, Chemistry; A Biology. **IGCSE** (2023): grade 9 in all nine subjects.

RESEARCH AND WORK EXPERIENCE

Engineering intern (incoming), Cryogenic Ltd, London *from Jan 2027*

- Joining the team that designs and builds cryogen-free superconducting magnets and low-temperature measurement systems, including the in-house electronics and control software.

Student researcher, surgical tool, with Rozenn Raffaut and Dr Agostino Stilli, UCL *Oct 2026 – present*

- Designing the next prototype of a 12 mm keyhole tool that grips a soft imaging skin and lays it onto an organ through a surgical port, through **19 CAD revisions in ten days** on the researcher's feedback.

Research collaborator, medical-image deep learning, with Dr J. W. Asare, UENR, Ghana *2026 – present*

- Working on deep-learning detection of oral cancer from clinical photographs; ran my duplicate-image audit on the group's public datasets and shared the results.

Healthcare Assistant (internship), Grand Mandalay Hospital, Mandalay, Myanmar *Jan 2025 – Jun 2025*

- Six months on the wards, taking routine observations and supporting nurses with patient care; seeing how clinical staff use monitoring equipment shaped my interest in biomedical devices.

PUBLICATIONS & PREPRINTS

Duplicate leakage in a public conjunctival pallor benchmark, and an honest baseline for image-based anaemia screening

Preprint. Htet, P. H. (2026). Zenodo. doi.org/10.5281/zenodo.22782147. Sole author. Code: github.com/Pai-88/pallor-hb

- Showed 710 records in a published clinical image dataset are really 383 photographs; the copies cross sites with conflicting labels, so site-grouped folds, the standard safeguard, still left two-fifths of a **+0.176** AUROC inflation.
- Corrected baseline: AUROC **0.706** (95% CI 0.653–0.757); every number regenerated by script and verified automatically (68 claims); code and detector released.

PROJECTS

PURSUIT-1: Person-Following Quadcopter *2026*

- Designed the 4-layer board that is also the airframe, from a Python netlist; rev A fabricated and assembled.
- Wrote the ESP32-S3 flight firmware: sensor fusion, cascaded PID, motor mixing and failsafes. Bench-tested, not yet flown.
- Camera pipeline for detection, tracking and face recognition: **0 false accepts in 24,110 impostor faces**; 369 tests.

KiCad and its Python API, ngspice, Python, C++, ONNX Runtime, OpenCV, PlatformIO. Code: github.com/Pai-88/pursuit-drone

EIT-16CH: Electrical Impedance Tomography Instrument *2026*

- Designed a 16-electrode 4-layer measurement board, 100 components, DRC clean; not yet fabricated.
- Compared learned and classical image reconstruction in simulation: a linear learned inverse wins only below 40 dB, a U-Net at every noise level tested (**3.3 times** the classical score at 20 dB).

KiCad, Freerouting, ESP32-S3, PlatformIO, PyTorch, NumPy, pyEIT.

TT-Scout: Table Tennis Match Analysis from Phone Video *2026*

- Finds every point and its winner in one phone video and measures 3D shot speed, calibrated from the table itself.
- Point winners correct on all **21** endings judged by eye, 5 of them blind; median speed error 3% on simulated shots.

Python, OpenCV, NumPy, PyTorch. Code: github.com/Pai-88/tt-scout

OXI-1: Fingertip Pulse Oximeter *2026*

- Designed a 55 × 28 mm 2-layer board carrying the optics, analogue front end and 16-bit ADC to fit inside a printed fingertip shell. Passed every release gate and three design reviews; boards now fabricated.
- Enforced the analogue constraints in code rather than by eye: a locked pre-route for the photodiode net, a release check on ground coverage under the front end, and a pick-and-place rotation table verified pin by pin against the fab's own footprints.

KiCad and its Python API, Python, CadQuery, ESP32-S3, PlatformIO.

Hand-Crank Phone Charger: Printed Planetary Drive, Stepper Motor as Alternator *2026*

- Designed and printed a 5.35:1 herringbone planetary stage, printed in place, that drives a NEMA23 stepper as an alternator.
- Pre-registered a SPICE prediction of **10 W** at the phone before building it, then measured **10.12 W** at 100 rpm on the finished machine.

Fusion 360, Blender, ngspice, Python; PETG on a Bambu P1S. Design document at paingheinhtet.com

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Git; PyTorch, scikit-learn, OpenCV, NumPy, pandas.
Machine learning: model evaluation and leakage control, calibration, uncertainty estimation, computer vision.
Electronics and signals: PCB design in KiCad, LTspice, ESP32 and RP2040 firmware, soldering; EMG and PPG signal processing.
CAD and lab: Fusion 360, SolidWorks, CadQuery, 3D printing (Bambu P1S); aseptic technique, cell passaging, Instron tensile testing.